

Order of Operations

BIM Boardroom + SSv3 + Fab — personal notebook edition

Generated from **ORDER_OF_OPERATIONS.md**. Every subsection includes a note pad; each major part adds full blank pages for handwriting.

USING THIS ON IPAD (APPLE PENCIL)

- **Best format:** PDF — import into **GoodNotes**, **Notability**, **Freeform**, or open in **Files** and tap Markup.
- **Get the PDF:** Open the `.html` sibling file in Safari or Edge → Share / Print → **Save as PDF** → AirDrop or iCloud to iPad.
- **Or:** Copy `ORDER_OF_OPERATIONS_NOTEBOOK.pdf` from the `docs/` folder if the build script created it on your PC.
- **Write on it:** Use Apple Pencil directly on the PDF; add extra blank pages in GoodNotes/Notability if you need more room.

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Order of Operations — BIM Boardroom + SSv3 + Fab

End-to-end operator sequence from Boardroom setup through shop Fab. Use this when the system feels big; the process itself is linear.

Contents: Part A (main nav dashboards) · Part B (project boards) · Part C (SSv3 → Fab) · **Part D (job levels & permissions — every detail)**

Flow diagram — see the full ORDER_OF_OPERATIONS.md in the repo for the mermaid chart.

NOTES — INTRODUCTION (SUMMARY)

Prerequisites

- **BIM Boardroom** running in Electron (`npm run electron:dev` or installed build) so the local API publishes projects/tasks.
- **Revit** with **Spooling Savant V3 (Exports)** loaded (SS Manager V3).

- Default export folder:

Spooling Savant Version 3 (Exports)/Boardroom/Exports

NOTES — PREREQUISITES (SUMMARY)

Part A — Dashboards (main nav)

Nav tabs come from `MainNav.tsx`. Who sees a gated tab is controlled by permissions (`permissions.ts`) plus Owner auto-grants. Dashboard **roles** (Shop Super, Warehouse Lead, etc.) are

assigned on **Employees** and shape *what you see inside* Fab / PM / Field / Shipping — they are not the same as the tab permission.

Quick map

Nav label	Tab id	Who sees the tab	What it is
Owner Dashboard	owner-dashboard	view-owner-dashboard / Owner	Portfolio health
PM Dashboard	pm-dashboard	view-pm-dashboard	PM's projects + PM board tasks
Clients	clients	Everyone	Main Dashboard + all project boards
Task Board	task-board	Everyone	Detailer / Support kanban
Fab Workstation	fab-dashboard	view-fab-dashboard	Queued / Warehouse / personal shop views
Shipping Dashboard	shipping-dashboard	view-shipping-dashboard	Shipping board task list
Field Dashboard	field-dashboard	view-field-dashboard	Field board task list
Time Tracking	time-tracking	view-time-tracking (default on for all job levels)	Hours calendar + Fab clocks
Employees	employees	Everyone (edit needs manage-org)	Roster, permissions, dashboard roles
Organizational Chart	org-chart	view-org-chart or manage-org	Reporting tree
Activity Log	activity-log	view-activity-log or Owner/BIM/Ops Mgr	Audit trail
Access Control	visibility-dashboard	Job-level setting (default Owner/BIM/Ops Mgr) or view-visibility-dashboard	Permissions + nav visibility matrix

NOTES — QUICK MAP

A1. Owner Dashboard

- **Purpose:** Executive portfolio view — how all projects are doing at a glance.
- **Who:** Owners (`view-owner-dashboard` / owner org category). BIM Manager does **not** get this by default.
- **Shows:** Active project count, average progress, budget vs spent, at-risk signals, BIM lifecycle stage rollup, full project list.
- **Actions:** Open a project into Clients → Main Overview. Read-only metrics.
- **Sub-modes:** None.
- **Linked boards:** Opens into Clients → Main Overview for a project.

NOTES — A1. OWNER DASHBOARD

A2. PM Dashboard

- **Purpose:** Project Managers' home for *their* jobs (or all projects if `manage-org`).
- **Who:** `view-pm-dashboard` — Owner; BIM / Ops / Support Manager defaults; seeded PMs.
- **Dashboard roles:** Project Managers, Assistant PMs (assigned on Employees).
- **Shows:** Project cards (budget, schedule, lifecycle %), PM board task list, lifecycle legend.
- **Actions:** Open **Project Management** board for a project; **Assign PM** (if `manage-org`).
- **Linked board:** `project-managers` under Clients → project.
- **Sub-modes:** None.

NOTES — A2. PM DASHBOARD

A large rectangular area with a dashed border and horizontal lines, intended for notes or a diagram.

A3. Clients (entry to all project boards)

- **Purpose:** Create clients/projects and do day-to-day spreadsheet board work.
- **Who:** Everyone signed in.
- **Sub-modes:**

1. **Main Dashboard** (logo) — portfolio cards by client; click into a project. 2. **Board mode** — strips:
Clients → Projects → Boards.

- **What lives here:** Every project board (see **Part B**). This is the spreadsheet home for BIM work; Fab shop day-to-day is usually **Fab Workstation**, not the Fab spreadsheet tab.

A4. Task Board

- **Purpose:** Office kanban of work assigned to people (not the project board grid).
- **Who:** Everyone.
- **Sub-modes:** Detailers | Support Specialists.
- **Shows:** Unassigned pool + one column per employee on that board.
- **Actions:** Drag tasks onto people; reorder by priority/due date; Task Board status visibility settings.
- **Related boards:** Surfaces tasks from Detailers / Deliverables / Spooling workflows that are assigned to those people.

NOTES — A4. TASK BOARD

- **Also via strip:** Shop Super, Owner, upstream managers.
- **Shows:** Packages in **Queued** or **Pulling Material** only.
- **Actions:** Open **Bill of Materials** PDF; status Queued → Pulling Material → Material Pulled; **Clock in/out**; photos/comments.
- **Gate:** After **Material Pulled**, package leaves the warehouse list.

A5c. Personal workstations (Dept Lead / Worker)

- **Default for:** Shop Dept Managers and Fab Workers (filtered to assignments).
- **Strip:** One pill per person (name + role), excluding Shop Super / Warehouse Lead.
- **Dept Lead:** Packages where they are Dept Lead; assign package and/or assemblies to workers; advance work; photos/comments; clock.
- **Worker:** Assigned packages/assemblies; sheet viewer; weld-log tap-fill; status progression; photos/comments; clock.
- **Typical status after material:** In Fab (Mech/Plmb/HVAC) → QA Review → Rework (optional) → Ready to Ship → Complete.

Fab shared tools

- Sheet / PDF viewer (assemblies, Spools Combined, export files).
- Interactive **Weld Log** (tap Date / Welder ID / Initials).
- **Photos / Comments** on the package.
- **Acting as** / View-as when the full strip is not shown.

NOTES — A5. FAB WORKSTATION (SHOP)

A6. Shipping Dashboard

- **Purpose:** Department view of **Shipping** board tasks.
- **Who:** `view-shipping-dashboard` — Owner; BIM / Ops / Support Manager; seeded Shipping staff.
- **Dashboard roles:** Shipping Manager, Workers.
- **Shows:** Task table (number, title, project, status).
- **Actions:** Review status here; create/edit on Clients → **Shipping** board.
- **Gap:** Fab **Ready to Ship** does **not** auto-promote into Shipping yet.
- **Sub-modes:** None.

NOTES — A6. SHIPPING DASHBOARD

A7. Field Dashboard

- **Purpose:** Department view of **Field** board tasks (site install).
- **Who:** `view-field-dashboard` — Owner; BIM / Ops / Support Manager; seeded Field staff (PMs often get Field too).
- **Dashboard roles:** Site Superintendents, Foremen, Crew Leads.
- **Shows:** Field board task table.
- **Actions:** Review progress; staffing via Employees → Field roles; task work on Clients → **Field** board.
- **Sub-modes:** None.

NOTES — A7. FIELD DASHBOARD

A8. Time Tracking

- **Purpose:** Log and review hours (office + Fab package clocks).
- **Who (tab):** Everyone. **Visible entries:** yourself; people who report to you; Owners see all.
- **Shows:** Entry form, Quick Tasks, calendar (**day / week / month**), **Clocked in** strip for open Fab clocks.
- **Actions:** Add/edit/delete completed entries (15-minute grid); clock out open Fab entries; pick employee if allowed.
- **Fab link:** Warehouse / personal **Clock in/out** writes time entries with package `taskId` / client / project.

NOTES — A8. TIME TRACKING

A10. Organizational Chart

- **Purpose:** Visual reporting tree.
- **Who:** `view-org-chart` or `manage-org` (edit needs `manage-org`).
- **Actions:** Pan/zoom; with `manage-org` , drag people under managers.
- **Fab note:** Warehouse Dashboard strip also opens for managers **upstream** of Shop Super or Warehouse Lead.

NOTES — A10. ORGANIZATIONAL CHART

A11. Activity Log

- **Purpose:** Audit trail (creates, updates, deletes, status/column changes).
- **Who:** `view-activity-log` or Owner / BIM Manager / Ops Manager.
- **Actions:** Filter / search; restore deleted columns if `manage-columns` .

NOTES — A11. ACTIVITY LOG

Related UI (not main nav)

UI	Where	Notes
Reports	Header	Dialog; scope follows active area
View As	Header	Switch perspective (incl. Fab roster)
Login	Pre-auth	Sign-in before any dashboard

NOTES — RELATED UI (NOT MAIN NAV)

NOTES — PART A — DASHBOARDS (MAIN NAV) (SUMMARY)

EXTRA NOTES — PART A — DASHBOARDS (MAIN NAV)

MORE NOTES — PART A — DASHBOARDS (MAIN NAV)

A large, empty rectangular box with rounded corners and a light blue background. It contains 15 horizontal lines for text entry, spaced evenly from top to bottom.

Part B — Project boards (Clients → board mode)

Under a project, the board strip is: **Main Overview** first, then sub-boards (reorderable), then any **custom** boards.

Default sub-board order: Project Management → RFI → Documents → Detailers → Deliverables → Spooling → Fabrication → Shipping → Field.

Main Overview vs sub-boards

	Main Overview	Sub-boards
Layout	All project tasks under section groups	One board's tasks (mirrors that section's hierarchy; RFI/Documents are flat)
Default sections	PM, RFI, Documents, Detailers, Deliverables, Spooling (+ custom). Not Fab / Shipping / Field by default	Own tab
Columns	Union of section columns + trailing Board column	Board-local; no Board column
Statuses	Per section branch (that board's workflow)	That board's status list
Trade/level groups	On Detailers / Deliverables / Spooling sections	Same hierarchy on those ghost boards

NOTES — MAIN OVERVIEW VS SUB-BOARDS

B2. Project Management (project-managers)

- **Purpose:** Project setup & coordination — contract through clash/IFC readiness.
- **Who:** PMs / Assistant PMs; linked from **PM Dashboard**.
- **Default statuses:**

1. Not Started 2. Contract Review 3. Kickoff Complete 4. BEP Approved 5. Model Setup 6. Clash Cycle Active 7. Clashes Resolved 8. IFC Ready 9. On Hold 10. Complete

- **Special:** Template **PM checklist** groups/tasks; Main Overview section; fixed Due Date (not the shared workflow due-date pack); no status auto-assign teams.
- **Actions:** Run kickoff/BEP/clash checklist work; assign PMs from dashboard when allowed.

NOTES — B2. PROJECT MANAGEMENT (`PROJECT-MANAGERS`)

B3. RFI (rfi)

- **Purpose:** Track RFIs waiting on response vs closed.
- **Who:** Support specialists (assignee pool = project support); PMs/detailers watch via Overview.
- **Default statuses (locked list — only these two ids):**
1. Waiting for Response 2. Complete
- **Special: Flat board** — default columns mainly title + status (no assignee/due/duration); Main Overview section; new tasks default to Waiting for Response. Status list cannot be expanded beyond those two ids (label/color only).
- **Actions:** Log RFIs; mark complete when answered.

NOTES — B3. RFI (RFI)

- **Actions:** Track document intake and model linkage.

NOTES — B4. DOCUMENTS (`DOCUMENTS`)

B5. Detailers (detailers)

- **Purpose:** 3D detailing / modeling through hangers, detailer QA, ready for coordination.
- **Who:** Detailers (assignee pool = project detailers). Also on **Task Board** → **Detailers**.
- **Default statuses:**

1. Not Started 2. Backgrounds Linked 3. Modeling (LOD 300) 4. Hangers & Supports 5. Detailer QA 6. Ready for Coordination 7. Rework 8. On Hold 9. Complete

- **Special:** Main Overview section; **trade-first** level/trade grouping (with Deliverables/Spooling); workflow due-date columns; shared Main custom columns.
- **Actions:** Model, QA, hand off toward coordination / deliverables.

NOTES — B5. DETAILERS (`DETAILERS`)

B6. Deliverables (deliverables)

- **Purpose:** Support Specialists pipeline — pre-planning through spool prep, with detailer handoffs.
- **Who:** Support (primary) + detailers on handoff statuses. **Task Board** → **Support Specialists**.
- **Default statuses (auto-assign team):**

1. Not Started → detailers 2. Ready for Pre-Planning → support 3. Pre-Planning Complete → support 4. Support In Progress → support 5. Ready for Spooling → support 6. Spool In Progress → support 7. Spool QA Review → support 8. Spool Approved → support 9. Ready for Fab → detailers 10. On Hold → detailers 11. Detailer Review → detailers 12. Fix Mark Ups → support 13. Complete → detailers

- **Special:** Status **auto-assign** to project detailer/support teams; assignees can be locked for overrides; trade-first groups; workflow due dates. **Ready for Fab** on Deliverables does not promote to Fab — promotion is **Spooling + SSv3** only.
- **Actions:** Pre-plan, support modeling, spool prep handoffs, mark-up loops.

NOTES — B6. DELIVERABLES (`DELIVERABLES`)

B7. Spooling (spooling)

- **Purpose:** Spool sheet creation, QA, approval, and the **fab-ready gate**.
- **Who:** Support / spoolers; detailers at Ready for Fab.
- **Default statuses (auto-assign team):**

1. Not Started → support 2. Ready for Spooling → support 3. Spool In Progress → support 4. Spool QA Review → support 5. Spool Approved → support 6. Ready for Fab → detailers 7. On Hold → support 8. Complete → support

- **Special:**
- **SSv3 export attach** — `boardroom-package.json` + reports land on the Spooling task (folder watch / import).
- Setting status to **Ready for Fab** with an SSv3 export **promotes** the package tree to the **Fab** board as **Queued**.
- After Ready for Fab / on Fab, wipe/replace of that export is **blocked**.
- Main Overview section; trade-first; workflow due dates.
- **Actions:** Run SSv3; attach export; QA sheets; flip **Ready for Fab** when shop should see it.
- **Keep Boardroom open** while SSv3 loads projects/tasks from `http://127.0.0.1:17321` .

NOTES — B7. SPOOLING (`SPOOLING`)

B8. Fabrication (fab)

- **Purpose:** Shop fabrication after promote — warehouse pull through dept fab, QA, ready to ship.
- **Who:** Fab shop (Shop Super, Warehouse, Dept Managers, Workers). Day-to-day UI is **Fab Workstation**; this tab is the Clients spreadsheet of the same tasks.
- **Default statuses:**

1. Not Started 2. Queued 3. Pulling Material 4. Material Pulled 5. In Fab (Mech) 6. In Fab (Plmb) 7. In Fab (HVAC) 8. QA Review 9. Rework 10. Ready to Ship 11. Complete
- **Special:** No default Main Overview section; receives SSv3 package/assembly trees on promote; warehouse filters Queued / Pulling Material; package custom fields (ssv3*); weld log / clock live in Fab Workstation. **Ready to Ship does not auto-create Shipping tasks.**
- **Actions:** Prefer Fab Workstation for assign, BOM, clock, sheets, weld log, photos/comments.

NOTES — B8. FABRICATION (`FAB`)

B9. Shipping (shipping)

- **Purpose:** Staging → loading → transit → site delivery → field receipt.
- **Who:** Shipping Manager / Shipping workers (**Shipping Dashboard**).
- **Default statuses:**

1. Not Started 2. Staging 3. Loading 4. In Transit 5. Delivered to Site 6. Received by Field 7. Complete

- **Special:** No default Main Overview section; workflow due dates; listed on Shipping Dashboard. No auto-promote from Fab yet.
- **Actions:** Create/advance shipping tasks; track delivery to field.

NOTES — B9. SHIPPING (`SHIPPING`)

B10. Field (field)

- **Purpose:** Site install — mobilization through punch, as-builts, final inspection.
- **Who:** Site supers, foremen, crew leads (**Field Dashboard**); PMs often watch.
- **Default statuses:**

1. Not Started 2. Mobilization 3. Material On Site 4. Rough-In 5. Hydro / Test 6. Trim-Out 7. Punch List 8. As-Built Update 9. Final Inspection 10. Complete
- **Special:** No default Main Overview section; workflow due dates; Field Dashboard lists these tasks.
- **Actions:** Track install progress on site.

NOTES — B10. FIELD (`FIELD`)

B11. Custom boards (`cb-*`)

- **Purpose:** Per-project extra tabs (user-named).
- **Who:** Whatever the team assigns (assignee pool often detailers+support when set).
- **Default statuses:** Generic Main list unless overridden in status settings.
- **Special:** Can appear as a Main Overview section; reorderable with built-ins.
- **Actions:** Same spreadsheet patterns as other hierarchical boards, project-defined workflow.

NOTES — B11. CUSTOM BOARDS (`CB-*`)

NOTES — PART B — PROJECT BOARDS (CLIENTS → BOARD MODE) (SUMMARY)

EXTRA NOTES — PART B — PROJECT BOARDS (CLIENTS → BOARD MODE)

MORE NOTES — PART B — PROJECT BOARDS (CLIENTS → BOARD MODE)

Part C — Order of operations (SSv3 → Fab)

1. Org / roles (Boardroom)

- Log in as Owner / BIM Manager / Ops as needed.

- Maintain **Employees** and **Org Chart**.
- Assign Fab dashboard roles:
- Shop Super
- Warehouse Lead + Warehouse Workers
- Shop Dept Managers (Mech / Plmb / HVAC)
- Fab Workers
- Optional: set each fab person's **Welder ID** for weld-log tap-fill.

NOTES — 1. ORG / ROLES (BOARDROOM)

5. Plot / Export to Boardroom (SSv3)

- Plot Packages / Export to Boardroom: pick Boardroom project + Spooling task + export folder.
- Writes `boardroom-package.json` + reports (Spools Combined, BOM, Cut List, Assembly List, Weld Log `.xlsx`, etc.).

NOTES — 5. PLOT / EXPORT TO BOARDROOM (SSV3)

6. Boardroom ingest

- Electron watches Exports; attaches package/assemblies/files to the Spooling task.
- Re-export can replace until Ready for Fab.

NOTES — 6. BOARDROOM INGEST

7. Ready for Fab (gate)

- Spooling status → **Ready for Fab**.
- Promotes package/assemblies to Fab as **Queued**.
- **Lock:** further SSv3 overwrite blocked.

NOTES — 7. READY FOR FAB (GATE)

11. Time Tracking

- Package clock in/out → Time Tracking tab.

NOTES — 11. TIME TRACKING

12. Shipping (later)

- Dashboard + board exist; no auto-hand-off from Ready to Ship yet.

NOTES — 12. SHIPPING (LATER)

NOTES — PART C — ORDER OF OPERATIONS (SSV3 → FAB) (SUMMARY)

Part D — Job levels, permissions & what each person sees

Two separate systems:

- 1. **Job level** (`orgCategory`) + some `role` rules → default **AppPermissions** → which **main nav tabs** appear.
- 2. **Dashboard roles** (Employees assigner: Shop Super, Warehouse Lead, PM, etc.) → how

Fab/PM/Field/Shipping behave **inside** a tab you already can open.

Permissions can also be edited per person on **Employees** (needs `manage-org`). Runtime auto-grants can still apply even if a chip is missing (see below).

D1. Permission catalog (every permission)

Permission id	Label in UI	What it unlocks
<code>edit-budget-hours</code>	Edit budget hours	Edit project budget hours. Also auto-unlocks View org chart at runtime.
<code>manage-org</code>	Manage org & permissions	Edit org chart layout, roster, permission chips, dashboard role assignments; on PM Dashboard see all projects + Assign PM. Alone also grants org-chart access .
<code>manage-columns</code>	Manage & restore columns	Delete/manage spreadsheet columns; restore deleted columns from Activity Log.
<code>view-activity-log</code>	View activity log	Activity Log main nav tab.
<code>view-org-chart</code>	View org chart	Organizational Chart main nav tab (browse). Drag/edit still needs <code>manage-org</code> .
<code>view-owner-dashboard</code>	View Owner dashboard	Owner Dashboard main nav tab.
<code>view-pm-dashboard</code>	View PM dashboard	PM Dashboard main nav tab.
<code>view-field-dashboard</code>	View Field dashboard	Field Dashboard main nav tab.
<code>view-fab-dashboard</code>	View Fab dashboard	Fab Workstation main nav tab.
<code>view-shipping-dashboard</code>	View Shipping dashboard	Shipping Dashboard main nav tab.
<code>view-visibility-dashboard</code>	View Access Control	Access Control main nav tab (permissions + visibility; also via editable job-level list).
<code>view-time-tracking</code>	View Time Tracking	Time Tracking main nav tab.

Always visible (no permission): Clients, Task Board, Employees. **Time Tracking rows** (who you can see): yourself, people who report up to you, or everyone if Owner — separate from the tab permission.

NOTES — D1. PERMISSION CATALOG (EVERY PERMISSION)

D2. Job levels (orgCategory)

Job level id	Label	Typical Employee.role
owner	Owner	Often support-specialist + owner category (Joe)
bim-manager	BIM Manager	support-specialist
operations-manager	Operations Manager	operations
operations-staff	Operations	operations (PM / Field / Fab / Shipping staff)
plumbing-detailer	Lead Plumbing Detailer	detailer
mechanical-detailer	Lead Mechanical Detailer	detailer
sheet-metal-detailer	Lead Sheet Metal Detailer	detailer
jr-detailer	Junior Detailer	detailer
support-manager	Support Specialist Manager	support-specialist
support-specialist	Support Specialist	support-specialist

If orgCategory is missing: operations → operations-staff ; support-specialist → support-specialist ; else → plumbing-detailer .

Employees UI stages (lanes, not permissions): Leadership | Detailers | Support | Project Management | Field | Fab Shop | Shipping.

NOTES — D2. JOB LEVELS (`ORGCATEGORY`)

D3. Default permissions by job level

Everyone starts with `view-org-chart` . Then category/role rules add more
(`createDefaultEmployeePermissions`).

Job level	Default permissions
Owner	All 10: Budget, Org, Columns, Activity, Chart, Owner, PM, Field, Fab, Shipping
BIM Manager	Budget, Org, Columns, Activity, Chart, PM, Field, Fab, Shipping — not Owner Dashboard
Operations Manager	Org, Columns, Activity, Chart, PM, Field, Fab, Shipping — not Budget, not Owner
Support Manager	Chart + PM + Field + Fab + Shipping
Detailer (any trade / junior)	Chart + Budget (<code>role === 'detailer'</code>)
Support Specialist	Chart only
Operations Staff (generic)	Chart only, plus roster extras below if seeded into a department

Operations staff roster extras (`operationsDashboardPermissions`)

Seeded roster	Example people	Extra view permissions
PM	Kendra, Damon	PM + Field + Fab + Shipping
Field	Marcus, Elena, Tyler	Field only
Fab	Gina...Priya (<code>emp-fab-1 ... 10</code>)	Fab only
Shipping	Harper, Mason	Shipping only

Runtime auto-grants (even if chip missing)

Permission / gate	Also true when
Edit budget hours	Detailer role or Owner or Joe Vasquez
View org chart	User can edit budget hours
Manage org	Owner or Joe Vasquez
View Owner dashboard	Owner
View PM / Field / Fab / Shipping	Owner
Manage columns	Explicit chip or Owner / BIM Manager / Ops Manager category
View activity log	Explicit chip or Owner / BIM Manager / Ops Manager category
View someone's time	Self or Owner or upstream manager in org chart

NOTES — D3. DEFAULT PERMISSIONS BY JOB LEVEL

D4. Main nav visibility by job level (defaults)

Always also: **Clients, Task Board, Employees.**

Job level	Owner	PM	Fab	Shipping	Field	Time	Org Chart	Activity	Access
Owner	✓	✓	✓	✓	✓	✓	✓	✓	✓
BIM Manager		✓	✓	✓	✓	✓	✓	✓	✓
Operations Manager		✓	✓	✓	✓	✓	✓	✓	✓
Support Manager		✓	✓	✓	✓	✓	✓		
Detailer (any)						✓	✓		
Support Specialist						✓	✓		
Ops – PM Lead		✓	✓	✓	✓	✓	✓		
Ops – PM Staff		✓				✓	✓		
Ops – Field Lead		✓			✓	✓	✓		
Ops – Field Staff					✓	✓	✓		
Ops – Fab Lead			✓	✓		✓	✓		
Ops – Fab Staff			✓			✓	✓		
Ops – Shipping Lead			✓	✓		✓	✓		
Ops – Shipping Staff				✓		✓	✓		
Ops – unassigned						✓	✓		

Employees tab: everyone can open; **editing** people/permissions/roles needs `manage-org` (Owner, BIM Manager, Ops Manager, Joe).

Access Control defaults to Owner / BIM Manager / Operations Manager and is editable on that dashboard (job-level checkboxes + Access column). All AppPermissions (Budget, Org, Columns, dashboards, etc.) are managed there — not on Employees.

NOTES — D4. MAIN NAV VISIBILITY BY JOB LEVEL (DEFAULTS)

D5. What each job level sees *inside* each dashboard

Owner

Surface	Detail
Owner Dashboard	Full portfolio metrics + all projects
PM Dashboard	All projects (<code>manage-org</code>); Assign PM
Clients / boards	Full spreadsheet access for all boards
Task Board	Detailers + Support kanban
Fab Workstation	Queued by default; full Dashboards strip (Queued, Warehouse, every personal tab)
Shipping / Field	Full department task lists
Time Tracking	All employees' entries
Employees	Full edit
Org Chart	View + edit
Activity Log	Full + restore columns

BIM Manager

Surface	Detail
Owner Dashboard	No tab
PM / Fab / Shipping / Field	Tabs yes; PM shows assigned projects unless also <code>manage-org</code>
Fab strip	Yes if upstream of Shop Super / Warehouse Lead in org chart (Ops Manager usually is; BIM Manager depends on reports-to)
Time Tracking	Self + reports
Employees / Org Chart	Edit (<code>manage-org</code>)
Activity Log	Yes

Operations Manager

Surface	Detail
Same phase dashboards as BIM Manager	PM, Fab, Shipping, Field, Org, Activity
Budget hours	No by default
Owner Dashboard	No
Fab strip	Yes when upstream of Shop Super / Warehouse Lead (seeded: Shop Super reports to Ops Manager)
Employees	Edit

Support Manager

Surface	Detail
PM / Fab / Shipping / Field	Tabs yes
Owner / Activity	No by default
Org Chart	View
Employees	View; edit only if given <code>manage-org</code>
Fab inside	Depends on fab role assignment + strip rules

Detailer (Plumbing / Mechanical / Sheet Metal / Junior)

Surface	Detail
Phase dashboards	None by default
Clients boards	Full project work — Detailers board primary; Main Overview sections
Task Board	Detailers lane
Budget hours	Yes
Org Chart	View
Time Tracking	Self (+ reports if any)

Support Specialist

Surface	Detail
Phase dashboards	None by default
Clients boards	Deliverables / Spooling / Documents / RFI primary
Task Board	Support Specialists lane
Org Chart	View

Operations – PM staff (e.g. Project Manager / Assistant PM roles)

Surface	Detail
Tabs	PM + Field + Fab + Shipping (+ Chart)
PM Dashboard	Projects where they are on <code>pmIds</code> (or all PM seed staff); not all projects unless <code>manage-org</code>
Field / Fab / Shipping	Tabs open; Fab mode still depends on fab role (usually none → queue fallback if they open Fab)
Dashboard roles	Project Managers / Assistant PMs buckets on Employees

Operations – Field staff

Surface	Detail
Tabs	Field (+ Chart)
Field Dashboard	All Field-board tasks (no per-role row filter)
Roles	Site Superintendent / Foreman / Crew Lead (staffing labels)

Operations – Fab staff

Surface	Detail
Tabs	Fab (+ Chart) only by default
Shop Super	Queued Dashboard; full strip
Warehouse Lead	Warehouse Dashboard; full strip
Warehouse Worker	Warehouse mode; no full strip
Dept Manager Mech/Plmb/HVAC	Personal workstation (packages as Dept Lead)
Worker	Personal workstation (package/assembly assignments)
Upstream of Super/Lead	Strip browse including Warehouse / Queued

Operations – Shipping staff

Surface	Detail
Tabs	Shipping (+ Chart)
Shipping Dashboard	All Shipping-board tasks
Roles	Shipping Manager / Workers

NOTES — D5. WHAT EACH JOB LEVEL SEES *INSIDE* EACH DASHBOARD

D6. Dashboard roles (Employees assigner) — not the same as job level

Assigned under Employees. They do **not** open the nav tab by themselves (you still need `view-*-dashboard`).

Dashboard	Role ids / labels
PM	<code>project-manager</code> Project Managers, <code>assistant-pm</code> Assistant PMs
Field	<code>site-superintendent</code> , <code>foreman</code> , <code>crew-lead</code>
Fab	<code>shop-super</code> , <code>warehouse-lead</code> , <code>warehouse-worker</code> , <code>dept-manager-mech</code> , <code>dept-manager-plmb</code> , <code>dept-manager-hvac</code> , <code>worker</code>
Shipping	<code>shipping-manager</code> , <code>worker</code>

Fab role → mode (inside Fab Workstation)

Primary role resolution order: Owner → Shop Super → Warehouse Lead → Warehouse Worker → Dept Manager (Mech/Plmb/HVAC) → Worker → else `owner-queue` (same as Queued).

Primary Fab role	Mode	Packages shown
Owner / Shop Super / <code>owner-queue</code> (no fab role)	Queued	All packages
Warehouse Lead / Warehouse Worker	Warehouse	Status Queued or Pulling Material only
Dept managers / Workers	Personal	Packages where user is Dept Lead, package Worker, or assembly assignee

Fab Dashboards strip (who)

Who	Strip?
Owner	Yes
Shop Super	Yes
Warehouse Lead	Yes
Anyone whose primary resolves to <code>owner-queue</code> (has Fab tab, no fab role)	Yes
Upstream manager of Shop Super or Warehouse Lead	Yes
Warehouse Worker / Dept Mgr / Worker (primary)	No (Acting-as / own mode only)

Fab in-mode actions

Action	Who can
Assign Dept Lead on a package	Queued mode + Shop Super or <code>owner-queue</code>
Assign Workers / assemblies	Personal mode + you are that package's Dept Lead
Clock in/out on package	Warehouse + personal workstations (feeds Time Tracking)
Browse Queued / Warehouse / every personal tab via strip	Strip-eligible users above

NOTES — D6. DASHBOARD ROLES (EMPLOYEES ASSIGNER) — NOT THE SAME AS JOB LEVEL

D7. Protected / special people

Person	Notes
Joe Vasquez (emp-support-1)	Owner category locked; cannot delete; full Owner permissions; explicit Budget + Manage org
Taylor Morgan (emp-owner-1)	Support Manager category locked; cannot delete; Support Manager dashboard defaults; reports to Joe

NOTES — D7. PROTECTED / SPECIAL PEOPLE

D8. Mental model

1. **Job level** → default **permissions** → **which tabs** you see. 2. **Dashboard role** → **queue / warehouse / personal** (and PM project filter). 3. **Org chart reporting** → time visibility + Fab strip for managers above Shop Super / Warehouse Lead. 4. Chips on Employees can always grant more (or less, except runtime Owner/detailer auto-grants).

NOTES — D8. MENTAL MODEL

NOTES — PART D — JOB LEVELS, PERMISSIONS & WHAT EACH PERSON SEES (SUMMARY)

EXTRA NOTES — PART D — JOB LEVELS, PERMISSIONS & WHAT EACH PERSON SEES

MORE NOTES — PART D — JOB LEVELS, PERMISSIONS & WHAT EACH PERSON SEES

SCRATCH PAD — PART D — JOB LEVELS, PERMISSIONS & WHAT EACH PERSON SEES

Status cheat sheet (compact)

Spooling

Not Started → Ready for Spooling → Spool In Progress → Spool QA Review → Spool Approved → **Ready for Fab** → On Hold / Complete

NOTES — SPOOLING

Fab

Warehouse-active only: **Queued, Pulling Material**

NOTES — STATUS CHEAT SHEET (COMPACT) (SUMMARY)

Who uses which Fab mode

Role	Mode
Owner / Shop Super	Queued (full Dashboards strip)
Warehouse Lead	Warehouse (strip too)
Ops / BIM managers above Shop Super or Warehouse Lead	Same strip
Warehouse Worker	Warehouse (own view; no full strip)
Shop Dept Manager (Mech / Plmb / HVAC)	Personal (Dept Lead packages)
Fab Worker	Personal (assignments)

NOTES — WHO USES WHICH FAB MODE (SUMMARY)

Key folders / URLs

What	Where
Boardroom API	<code>http://127.0.0.1:17321</code>
Default SSv3 export folder	<code>Spooling Savant Version 3 (Exports)/Boardroom/Exports</code>
Package manifest	<code>boardroom-package.json</code> in that folder
SSv3 add-in notes	[<code>Spooling Savant Version 3 (Exports)/README.md</code>] (<code>./Spooling%20Savant%20Version%203%20(Exports)/README.md</code>)

NOTES — KEY FOLDERS / URLS (SUMMARY)

GENERAL NOTES

GENERAL NOTES (CONTINUED)

